

SQL Query Filtering Lab

Filtering SQL Queries for Cybersecurity Investigations

Overview

In this lab, I applied SQL filtering techniques within the MariaDB shell to retrieve targeted security-related information from the organization database. The investigation focused on employee devices, operating systems, department information, and office locations to support security operations and organizational asset management.

This project demonstrates practical cybersecurity data analysis using SQL filters such as WHERE and LIKE to efficiently locate specific records within a database.

Objectives

- Retrieve organization machine information
- Filter devices by operating system
- Identify employees in specific departments
- Investigate employee office locations
- Use SQL filtering operators such as WHERE and LIKE
- Strengthen SQL investigation and database querying skills

Environment

Component	Details
Database	MariaDB
Database Name	Organization
Tables Used	machines, employees
Operating Environment	Linux Bash Shell

Task 1 – List All Organization Machines

Retrieve Device IDs and Operating Systems

SQL Query

```
SELECT device_id, operating_system  
FROM machines;
```

```
MariaDB [organization]> SELECT device_id, operating_system FROM machines;  
+-----+-----+  
| device_id | operating_system |  
+-----+-----+  
| a184b775c707 | OS 1 |  
| a192b174c940 | OS 2 |  
| a305b818c708 | OS 3 |  
| a317b635c465 | OS 1 |  
| a320b137c219 | OS 2 |  
| a398b471c573 | OS 3 |  
| a667b270c984 | OS 1 |  
| a821b452c176 | OS 2 |  
| a998b568c863 | OS 3 |  
| b157c491d493 | OS 2 |  
| b239c825d303 | OS 1 |  
| b264c773d977 | OS 2 |  
| b265c937d713 | OS 2 |  
| b433c245d868 | OS 1 |  
| b551c837d758 | OS 3 |  
| b566c710d544 | OS 1 |  
| b806c503d354 | OS 2 |  
| b979c871d361 | OS 2 |  
| c116d593e558 | OS 3 |  
| c150d982e144 | OS 2 |  
| c185d679e493 | OS 1 |  
| c406d877e950 | OS 2 |  
| c547d140e477 | OS 2 |  
| c568d742e974 | OS 2 |  
| c597d792e215 | OS 2 |  
| c603d749e374 | OS 1 |  
| c986d200e170 | OS 2 |  
| d168e758f876 | OS 2 |  
| d280e557f635 | OS 3 |  
| d336e475f676 | OS 2 |  
| d394e816f943 | OS 3 |  
| d647e310f618 | OS 2 |
```

Purpose

Retrieved device identifiers and operating systems for all organizational machines to support asset visibility and security update planning.

Analysis

The query returned 200 organizational machine records, indicating the total number of systems tracked within the database inventory.

Task 2 – Retrieve Machines Running OS 2

Filter Machines by Operating System

SQL Query

```
SELECT device_id, operating_system  
FROM machines  
WHERE operating_system = 'OS 2';
```

```
MariaDB [organization]> SELECT device_id operating_system  
-> FROM machines  
-> WHERE operating_system = 'OS 2';  
-----+  
operating_system |  
-----+  
a192b174c940     |  
a320b137c219     |  
a821b452c176     |  
b157c491d493     |  
b264c773d977     |  
b265c937d713     |  
b806c503d354     |  
b979c871d361     |  
c150d982e144     |  
c406d877e950     |  
c547d140e477     |  
c568d742e974     |  
c597d792e215     |  
c986d200e170     |  
d168e758f876     |  
d336e475f676     |  
d647e310f618     |  
d693e351f221     |  
e113f288g203     |  
e218f877g788     |  
e395f616g566     |  
f229g533h679     |  
f370g535h632     |  
f934g229h883     |  
g299h520i457     |  
g770h829i938     |  
g773h303i639     |  
g950h972i991     |  
g953h643i618     |  
h174i497j413     |  
h835i179i862     |
```

Purpose

Identified all machines running `OS 2` so the security team could prioritize system updates and vulnerability remediation.

Analysis

A total of 80 systems were identified running 'OS 2', which allowed the security team to prioritize operating system updates and remediation efforts.

Task 3 – List Employees in Specific Departments

Retrieve Employees in the Finance Department

SQL Query

```
SELECT *  
FROM employees  
WHERE department = 'Finance';
```

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Finance';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1010	k242l212m542	jlansky	Finance	South-109
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1029	d336e475f676	ivelasco	Finance	East-156
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1062	k367l639m697	redwards	Finance	North-180
1069	NULL	jpark	Finance	East-110
1076	y347z204a710	fgarcia	Finance	Central-270
1081	d647e310f618	qcorbit	Finance	South-290
1083	f840g812h544	gkoshi	Finance	West-165
1099	v283w690x104	anaser	Finance	West-357
1105	b551c837d758	kmei	Finance	Central-232
1122	s103t952u851	btorres	Finance	West-319
1136	g299h520i457	jhawes	Finance	West-416
1138	i671j355k725	sromero	Finance	South-329

Purpose

Retrieved employee information from the Finance department to support privacy notice distribution and handling of confidential financial information.

Analysis

The first employee record returned from the Sales department query belonged to employee ID '1003'.

Retrieve Employees in the Sales Department

SQL Query

```
SELECT *  
FROM employees  
WHERE department = 'Sales';
```

```
MariaDB [organization]> SELECT *  
-> FROM employees  
-> WHERE department = 'Sales';
```

employee_id	device_id	username	department	office
1009	NULL	lrodriqu	Sales	South-134
1011	1748m120n401	drosas	Sales	South-292
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1057	f370g535h632	mscott	Sales	South-270
1063	l686m140n569	lpope	Sales	East-226
1066	o678p794q957	ttyrell	Sales	Central-444
1071	t244u829v723	zdutchma	Sales	West-348
1072	u905v920w694	esmith	Sales	East-421
1078	a667b270c984	sharley	Sales	North-418
1085	h339i498j269	cperez	Sales	East-325
1086	i281j129k749	lmajumda	Sales	West-499
1089	l358m929n154	jpark2	Sales	West-251
1091	n378o313p469	rtran	Sales	Central-230
1092	o391p779q935	lpark	Sales	West-227
1098	u671v146w618	tarchamb	Sales	North-423
1107	d168e758f876	akajwara	Sales	North-471
1109	f229g533h679	nlocklea	Sales	East-196
1110	g567h376i314	pchaudhu	Sales	Central-428
1111	h835i179j862	jlee	Sales	West-309
1116	m272n572o874	nzhao	Sales	South-100
1117	n683o758p820	dahmad	Sales	West-405
1118	o305p208q337	jpark3	Sales	South-329
1119	p164q780r999	omubarak	Sales	West-409

Purpose

Retrieved employee information from the Sales department for department-specific communication and administrative tasks.

Analysis

The query identified 33 employees assigned to the Sales department.

Task 4 – Identify Employee Machines

Identify the Employee Using Office South-109

SQL Query

```
SELECT *  
FROM employees  
WHERE office = 'South-109';
```

```
MariaDB [organization]> SELECT *  
  -> FROM employees  
  -> WHERE office = 'South-109';  
+-----+-----+-----+-----+-----+  
| employee_id | device_id   | username | department | office   |  
+-----+-----+-----+-----+-----+  
|          1010 | k242l212m542 | jlansky  | Finance    | South-109 |  
+-----+-----+-----+-----+-----+  
1 row in set (0.001 sec)
```

Purpose

Identified the employee assigned to office 'South-109' so the security team could send an alert regarding a machine issue.

Analysis

The employee assigned to the affected machine located in office 'South-109' was identified as 'jlansky' from the Finance department.

Retrieve All Employees in the South Building

SQL Query

```
SELECT *  
FROM employees  
WHERE office LIKE 'South%';
```

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE office LIKE 'South%';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1013	n205o559p243	zbernal	Information Technology	South-229
1020	u899v381w363	arutley	Marketing	South-351
1021	v200w121x977	smartell	Information Technology	South-138
1024	y976z753a267	iuduike	Sales	South-215
1035	j236k303l245	bisles	Sales	South-171
1048	w167x592y375	tmitchel	Finance	South-288
1057	f370g535h632	mscott	Sales	South-270
1058	g264h852i697	madebowa	Marketing	South-119
1059	h832i322j795	jnguyen	Marketing	South-255
1064	NULL	ejones	Marketing	South-477
1081	d647e310f618	qcorbit	Finance	South-290
1084	g950h972i991	nhuynh	Human Resources	South-155
1090	m891n748o375	cpatel	Information Technology	South-351
1097	t363u179v368	jlee	Human Resources	South-254
1102	y943z930a241	kselassi	Marketing	South-378
1106	c597d792e215	jcohen	Marketing	South-395
1112	i772j807k175	ccruz	Information Technology	South-298
1116	m272n572o874	nzhao	Sales	South-100
1118	o305p208q337	jpark3	Sales	South-329
1123	t479u468v151	ekonya	Human Resources	South-445
1125	v491w553x421	mrodgers	Marketing	South-490
1131	b265c937d713	asierra	Information Technology	South-159
1137	h165i539j638	mwood	Human Resources	South-166

Purpose

Retrieved all employees located in the South building to investigate organization-wide issues affecting multiple machines in that location.

Analysis

The first employee returned from the South building query belonged to the Finance department.

Investigation Summary

The SQL filtering investigation revealed:

- Organizational machines operating on multiple operating systems
- Devices requiring updates based on operating system versions
- Department-specific employee information for administrative and security notifications
- Office-based employee identification for incident response activities
- Efficient use of SQL filtering techniques to narrow investigation scope
- Total organizational machine inventory consisted of 200 systems

- 80 systems were identified running 'OS 2'
- 33 employees were identified in the Sales department
- Affected employee systems were successfully identified for incident response follow-up

These findings demonstrate how SQL filters can improve the speed and accuracy of cybersecurity investigations and organizational asset management.

Skills Demonstrated

- SQL filtering with 'WHERE'
- Pattern matching using 'LIKE'
- Database investigation techniques
- Security asset management
- Employee and device identification
- Data retrieval and filtering
- MariaDB command-line usage
- Operating system investigation
- Security incident response support
- SQL-based asset tracking

Cybersecurity Relevance

Security analysts frequently use SQL filters to locate systems, users, and security events relevant to ongoing investigations. Efficient filtering reduces investigation time and improves operational visibility.

These skills are critical for:

- Threat hunting
- Incident response
- Vulnerability management
- Asset inventory management
- Security operations
- Compliance investigations

This lab strengthened practical experience in filtering SQL queries for cybersecurity analysis and operational investigations.

Key Takeaways

- Learned how to apply filters to SQL queries using 'WHERE'
- Practiced narrowing search results to retrieve relevant information efficiently
- Used the 'LIKE' operator for pattern matching in database investigations
- Improved SQL investigation workflows for cybersecurity operations
- Gained hands-on experience with MariaDB database filtering techniques

Conclusion

This lab provided practical experience in filtering SQL queries to retrieve targeted security and organizational data. By using SQL filtering operators such as 'WHERE' and 'LIKE', I improved my ability to efficiently investigate systems, employees, and office locations within a database environment.

These foundational SQL investigation skills are essential for cybersecurity analysts involved in monitoring, incident response, asset management, and security operations.

Author

Cornelius Donkor

Cybersecurity Professional | Security Analyst | SQL & Linux Enthusiast

